

Hole Cleaning & ECD Management in Deepwater (illustrative)

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1. Scope

Applies to large-diameter deepwater hole sections (12-1/4 in and 17-1/2 in) where annular fluid velocities must balance hole cleaning against excessive Equivalent Circulating Density (ECD).

2. Hydraulic Limits for 12-1/4 in Section across the U3 sand (4,195–4,290 m MD)

- Nominal flow rate: 850 gpm.
- Maximum allowable ECD: 12.00 ppg (shoe FIT: 12.10 ppg).
- Dynamic annular pressure loss must not exceed 0.35 ppg equivalent over static mud weight.

3. High-ROP Cuttings Management

- When ROP exceeds 25 m/hr, cuttings concentration in the annulus escalates rapidly, elevating ECD by up to 0.40 ppg.
- High-viscosity low-shear sweeps (20 bbl tandem pills) must be pumped every 100 m drilled.
- If ECD approaches within 0.10 ppg of fracture limit (12.00 ppg), immediately reduce flow rate or cap ROP.